

1 PURPOSE

This Work Instruction describes the determination of water content in solid and liquid samples by coulometric Karl Fischer titration.

2 SCOPE

Applies to raw materials, in-process samples and finished product samples.

3 RESPONSIBILITIES

The analyst performs the test. A second qualified analyst reviews the result. Calculated values may not be overridden by the analyst.

4 EQUIPMENT

Coulometric Karl Fischer titrator, model KF-1000. Analytical balance, 0.01 mg readability. Drying oven capable of 105 C.

5 REAGENTS

Karl Fischer anolyte reagent. Karl Fischer catholyte reagent. Sodium tartrate dihydrate certified reference standard. Anhydrous methanol.

6 SAFETY

WARNING: Karl Fischer reagents contain methanol and iodine. Wear nitrile gloves, safety goggles and a lab coat. Work in a fume hood. Do not breathe vapours.

7 PROCEDURE

7.1 Condition the titration cell until the drift is stable.

7.2 Record the cell drift in ug/min.

7.3 Weigh the sample directly into the titration vessel.

7.4 Start the titration and record the water found in ug.

7.5 Repeat for a total of three replicates.

8 PARAMETERS

Cell drift, unit ug/min, numeric, format 0.0, 1 replicate, mandatory yes.

Sample mass, unit mg, numeric, format 0.0000, 3 replicates, mandatory yes.

Water found, unit ug, numeric, format 0.0, 3 replicates, mandatory yes.

Water content, unit percent, calculated, format 0.000, 3 replicates, mandatory yes.

Mean water content, unit percent, calculated as the average of the replicates.

RSD of water content, unit percent, calculated as the relative standard deviation.

Analyst comment, text, optional.

9 CALCULATIONS

9.1 Water content percent equals water found in ug divided by sample mass in mg, divided by ten thousand, multiplied by one hundred.

9.2 The mean is the arithmetic average of the three replicate water content values.

9.3 The RSD is the standard deviation of the replicates divided by the mean, multiplied by one hundred.

9.4 The product specific release limit, where one exists, is read from the product master record and overrides the default specification.

10 REPORTING

Report the mean water content rounded to three decimal places.

11 LIMITS

System suitability: cell drift must be less than or equal to 15.0 ug/min. Not OOS generating.

System suitability: RSD of water content must be less than or equal to 2.0 percent. Not OOS generating.

Release limit: mean water content must be greater than or equal to 0.100 percent and less than or equal to 0.500 percent. OOS generating yes.
Alert limit: mean water content greater than 0.450 percent triggers a warning. Not OOS generating.